Bivariate Statistics — Paper 2

Practice Paper (calculator required)

Name: _____ Date: _____

Time allowed: 35 minutes **Maximum mark:** 30

Instructions. Answer all questions. A graphic display calculator is required. Unless otherwise stated in the question, give numerical answers either exactly or correct to three significant figures. Show your working: marks may be given for a correct method even if the answer is wrong.

Question 1 [Maximum mark: 10]

A cafe owner records the maximum daily temperature x (°C) and the number of iced drinks sold y on eight days.

x (°C)	14	17	19	22	24	27	29	32
y (drinks)	12	18	21	30	33	41	45	52

(a) Write down the value of r, Pearson's product-moment correlation coefficient. **[1]**

(b) Describe the correlation between x and y. **[2]**

(c) Find the equation of the regression line of y on x. **[2]**

(d) Interpret the gradient of the regression line. Explain why the y-intercept has no sensible meaning in this context. **[2]**

(e) Use the regression line to estimate the number of iced drinks sold when

(i) the maximum temperature is 25 °C, **[1]**

(ii) the maximum temperature is 40 °C. **[1]**

Comment on the reliability of each estimate. **[1]**

Question 2 **[Maximum mark: 10]**

A teacher records the number of hours x that ten students practised a task and their score y on a test. The score of the tenth student was not recorded.

x	4	6	7	9	10	12	13	15	16	**11**
y	18	23	22	31	30	38	41	47	50	**?**

(a) Using only the nine students with complete data, find the mean point $(\bar{x}, \bar{y})$. **[2]**

(b) Find the equation of the regression line of y on x for these nine students. **[2]**

(c) Show that the mean point lies on the regression line. **[2]**

(d) Estimate the score of the tenth student. **[2]**

(e) Comment on the reliability of your estimate, referring to r and to the range of the data. **[2]**

Question 3 **[Maximum mark: 10]**

A company records its advertising spend x (thousands of dollars) and its sales y (thousands of dollars) for ten months.

x	2	3	4	5	6	6	7	8	9	10
y	14	19	21	27	62	31	33	40	43	50

(a) Find r for all ten months, and describe the correlation. **[2]**

(b) Identify the outlier. Give a reason for your answer. **[1]**

(c) The outlier is removed. Find the new value of r and the equation of the new regression line of y on x. **[3]**

(d) The company says the outlier was a month with a one-off promotion that really did happen. State, with a reason, whether the outlier should be removed from the data. **[2]**

(e) Using your regression line from part **(c)** (the line for the nine months without the outlier), estimate the sales when the advertising spend is \$12 000, and comment on the reliability of your estimate. **[2]**

END OF PAPER